

Analytical Resistivity Jacobian for a 1D Layered VTI Electromagnetic Medium

Wouter Deleersnyder

Public Version

September 14, 2026

This document presents the full analytical derivation of the Jacobian for resistivity. Derivations for the remaining model parameters are intentionally not included here and are not planned for public release. This material is shared only upon individual request, in order to retain some control over how and by whom it is used.

If you are interested in the derivations for the other parameters, or in discussing potential applications, extensions, or collaboration, please feel free to reach out to me at

www.wouterdeleersnyder.be

Contents

1	Introduction	3
1.1	Scope and how to read this document	3
1.2	Layered VTI model and constitutive relations	3
1.2.1	Geometry	3
1.2.2	Constitutive parameters	3
1.2.3	TM/TE propagation constants	4
1.3	TM/TE decomposition, reflection coefficients, propagators	4
1.3.1	Local reflection coefficients	4
1.3.2	Global reflection coefficients	5
1.3.3	Source-layer field propagators and amplitude factors	5
2	Gradients with respect to resistivity/conductivity	6
2.1	Introduction	6
2.2	EM field derivatives	6
2.2.1	Derivatives of the local reflection coefficients	7
2.2.2	Derivatives of the global reflection coefficients	8
2.2.3	Derivatives of the EM field components	9
2.2.4	Derivatives of the integral coefficients	10
2.2.5	From σ -derivatives to a resistivity Jacobian	15
3	Other gradients	16
3.1	Gradients with respect to the anisotropy factor (aniso)	16
3.2	Gradients with respect to electric permittivity (epermH, epermV)	16
3.3	Gradients with respect to magnetic permeability (mpermH, mpermV)	16
3.4	Gradients with respect to interface depths (depth, thickness)	16
3.5	Gradients with respect to source/receiver depth	16
3.6	Gradients with respect to offset (off)	16
3.7	Gradients with respect to frequency (freq)	16
3.8	Gradients with respect to horizontal source position	16
3.9	Gradients with respect to source/receiver orientation	16
A	Notation	17
A.1	Layer/medium parameters	17
A.2	Propagation, reflection and kernel quantities	17

Chapter 1

Introduction

1.1 Scope and how to read this document

The electromagnetic field radiated by a dipole source in horizontally layered, homogeneous media can be expressed in the wavenumber domain, following a Hankel transform, in terms of up- and down-going TM- and TE-mode reflection coefficients and a small set of source-layer propagation factors. Differentiating this closed-form expression with respect to the conductivity of a given layer yields the conductivity sensitivity analytically, eliminating the need for finite-difference perturbations of the forward model.

Section 1.2 establishes the notation and presents the constitutive relations linking conductivity to the two complex per-layer quantities—admittivity and impedivity—used by the forward model. Section 1.3 introduces the reflection-coefficient and propagator framework before differentiation, which is then developed analytically in Chapter 2. The main result, presented in Chapter 2, is the resistivity Jacobian. Appendix A provides a consolidated reference for the notation and symbols used throughout.

Sign convention: $e^{+i\omega t}$ is used throughout, so that $s := i\omega$ with $\omega = 2\pi f$ the angular frequency.

1.2 Layered VTI model and constitutive relations

1.2.1 Geometry

The subsurface is represented by N horizontal, homogeneous VTI (vertical transverse isotropic) layers separated by planar interfaces at $z_1 < z_2 < \dots < z_{N-1}$. The upper and lower half-spaces are defined by $z_0 = -\infty$ and $z_N = +\infty$, respectively. Layer n extends over $z_{n-1} < z < z_n$ and, for an interior layer, has thickness $d_n = z_n - z_{n-1}$. The source is located at depth z_0^{src} in layer s , while the receiver is positioned at depth z .

1.2.2 Constitutive parameters

Each layer n is described by real, frequency-independent material properties: the horizontal and vertical resistivities ρ_n, ρ_n^v (equivalently the conductivities), the horizontal and vertical relative electric permittivities $\varepsilon_n, \varepsilon_n^v$, and the horizontal and vertical relative

magnetic permeabilities μ_n, μ_n^v . From these quantities, four complex frequency-dependent *admittivity/impedivity* parameters are defined, with $s := i\omega$:

$$\eta_n = \sigma_n + s \varepsilon_n = \frac{1}{\rho_n} + i\omega \varepsilon_0 \varepsilon_n, \quad \eta_n^v = \sigma_n^v + s \varepsilon_n^v = \frac{1}{\rho_n \lambda_n^2} + i\omega \varepsilon_0 \varepsilon_n^v, \quad (1.1)$$

$$\zeta_n = s \mu_n = i\omega \mu_0 \mu_n, \quad \zeta_n^v = s \mu_n^v = i\omega \mu_0 \mu_n^v, \quad (1.2)$$

Here $\lambda_n = \sigma_n / \sigma_n^v = \sqrt{\rho_n^v / \rho_n} \geq 1$ denotes the dimensionless *anisotropy factor*; ε_0 and μ_0 are the vacuum permittivity and permeability. The quantities η_n, η_n^v are referred to as *admittivities* and govern the electric response, whereas ζ_n, ζ_n^v are *impedivities* associated with the magnetic response. Together they form the *eta/zeta seeds* from which the forward model and its derivatives are constructed. For the conductivity derivatives in Chapter 2, only $\partial \eta_n / \partial \sigma_n = 1$ and $\partial \eta_n^v / \partial \sigma_n^v = 1$ are required, with the remaining constitutive parameters kept fixed.

1.2.3 TM/TE propagation constants

For a planar layered medium, Maxwell's equations can be separated into transverse-magnetic (TM) and transverse-electric (TE) modes. The TM mode is associated with the vertical electric field, whereas the TE mode is associated with the vertical magnetic field. Using the horizontal wavenumber κ as the Hankel-transform integration variable, define

$$\gamma_n = \sqrt{\zeta_n \eta_n^v}, \quad \bar{\gamma}_n = \sqrt{\eta_n \zeta_n^v}, \quad (1.3)$$

$$\Gamma_n = \left(\frac{\eta_n}{\eta_n^v} \right)^{1/2} (\kappa^2 + \gamma_n^2)^{1/2}, \quad \bar{\Gamma}_n = \left(\frac{\zeta_n}{\zeta_n^v} \right)^{1/2} (\kappa^2 + \bar{\gamma}_n^2)^{1/2}. \quad (1.4)$$

Γ_n and $\bar{\Gamma}_n$ are the TM and TE propagation constants, respectively. In the isotropic limit, $\eta_n = \eta_n^v$ and $\zeta_n = \zeta_n^v$, both reduce to the standard diffusion expression $\sqrt{\kappa^2 + s \mu_n \eta_n}$. The two modes, however, have different dependencies on the electric properties. $\bar{\gamma}_n$ contains the horizontal quantity η_n but not η_n^v , whereas Γ_n depends on both through γ_n and the factor $(\eta_n / \eta_n^v)^{1/2}$. Consequently, the TE mode is independent of vertical conductivity, $\partial \bar{\Gamma}_n / \partial \sigma_n^v \equiv 0$, as shown in Chapter 2, (2.21), while the TM mode remains sensitive to both horizontal and vertical properties.

1.3 TM/TE decomposition, reflection coefficients, propagators

This section summarizes the *primal*, non-Jacobian quantities that are differentiated in Chapter 2. These consist of the local and global reflection coefficients, the source-layer field propagators, and the TM/TE amplitude factors used to assemble the wavenumber-domain Green's functions.

1.3.1 Local reflection coefficients

At the interface separating layers n and $n+1$, the corresponding *local* single-interface reflection coefficients for the TE and TM modes are

$$r_n^{TE} = \frac{\bar{\Gamma}_n - \bar{\Gamma}_{n+1}}{\bar{\Gamma}_n + \bar{\Gamma}_{n+1}}, \quad r_n^{TM} = \frac{\eta_{n+1} \Gamma_n - \eta_n \Gamma_{n+1}}{\eta_{n+1} \Gamma_n + \eta_n \Gamma_{n+1}}. \quad (1.5)$$

These expressions follow from enforcing continuity of the tangential fields across the interface. The TE coefficient has the usual symmetric mismatch form, while the TM coefficient contains the additional η_{n+1}/η_n weighting because its characteristic impedance depends on Γ_n/η_n rather than on Γ_n alone.

1.3.2 Global reflection coefficients

Multiple reflections within the layered medium are incorporated using the standard recursive Bremmer-series formulation. The global reflection coefficient $R_n^{TE, TM}$, evaluated from the top of layer $n + 1$, is then given by

$$R_n^{TE, TM} = \frac{r_n^{TE, TM} + R_{n+1}^{TE, TM} E_{n+1}}{1 + r_n^{TE, TM} R_{n+1}^{TE, TM} E_{n+1}}, \quad E_{n+1} = \exp(-2\Gamma_{n+1}^{(\cdot)} d_{n+1}), \quad (1.6)$$

Here $\Gamma^{(\cdot)}$ denotes Γ for TM and $\bar{\Gamma}$ for TE. The recursion starts from the bottom half-space with $R_N = 0$ and proceeds upward; the analogous construction is used for the opposite propagation direction. Since the contribution of each layer contains the exponentially decaying factor E_{n+1} , deeper interfaces are increasingly attenuated. This is the physical basis for treating a conductivity perturbation as a local modification that is propagated outward through a sequence of $\partial R_k / \partial R_{k+1}$ factors (Chapter 2, (2.27), (2.33)).

1.3.3 Source-layer field propagators and amplitude factors

Inside source layer s , the vertical field dependence is represented by two exponential propagators referenced to the receiver depth z and the bounding interfaces z_{s-1} and z_s :

$$W_s^u = \exp[-\Gamma_s(z_s - z)], \quad W_s^d = \exp[-\Gamma_s(z - z_{s-1})], \quad (1.7)$$

The TE propagators $\bar{W}_s^{u,d}$ have the same form after replacing Γ_s by $\bar{\Gamma}_s$. The upward- and downward-going amplitude factors combine the source-layer global reflection coefficients R_s^- and R_s^+ with the multiple-reflection denominator

$$M_s = 1 - R_s^- R_s^+ \exp(-2\Gamma_s d_s) \quad (1.8)$$

are

$$P_s^{u\pm} = \frac{R_s^+}{M_s} \left(\exp(-\Gamma_s d^+) \pm R_s^- \exp[-\Gamma_s(d_s + d^-)] \right), \quad (1.9)$$

$$P_s^{d\pm} = \frac{R_s^-}{M_s} \left(\exp(-\Gamma_s d^-) \pm R_s^+ \exp[-\Gamma_s(d_s + d^+)] \right), \quad (1.10)$$

where $d^+ = z_s - z_0^{\text{src}}$ and $d^- = z_0^{\text{src}} - z_{s-1}$. The corresponding TE factors $\bar{P}_s^{u\pm}, \bar{P}_s^{d\pm}$ are obtained using $R^\pm \rightarrow R^{TE, \pm}$ and $\Gamma_s \rightarrow \bar{\Gamma}_s$. The TM and TE integral kernels entering the wavenumber-domain Green's functions (Chapter 2, (2.41), (2.75), (2.84)) are assembled from these building blocks:

$$\tilde{g}_{zh;s}^{TM} = P_s^{u-} W_s^u - P_s^{d-} W_s^d, \quad \tilde{g}_{hh;s}^{TM} = P_s^{u+} W_s^u + P_s^{d+} W_s^d, \quad \tilde{g}_{zz;s}^{TE} = \bar{P}_s^{u+} \bar{W}_s^u + \bar{P}_s^{d+} \bar{W}_s^d.$$

Chapter 2

Gradients with respect to resistivity/conductivity

Literature. Streich et al. (2011), *Sensitivity of controlled-source electromagnetic fields in planarly layered media*; Patriarca et al. (2011), *Gradient based technique for electromagnetic layered earth model data inversion*.

2.1 Introduction

Conductivity is commonly treated as either isotropic or VTI-anisotropic, in which case the horizontal and vertical conductivities are different (Everett and Constable 1999; Newman et al. 2010). As with the EM field components themselves, the corresponding derivatives can be written as Bessel-function integrals. These integrals can be evaluated efficiently with fast Hankel-transform digital filters (e.g. Johansen and Sørensen 1979; Anderson 1982; Christensen 1990), so that the computational effort required for sensitivities remains comparable to that of the forward-field calculation.

For a planar layered medium, Maxwell’s equations likewise allow the EM field to be separated into TE and TM contributions (Section 1.3).

2.2 EM field derivatives

For an arbitrary field component F , the sensitivity J_n^F is defined as the derivative with respect to the conductivity of layer n ,

$$J_n^F = \frac{\partial F}{\partial \sigma_n} \quad (2.1)$$

Alternatively, derivatives may be taken with respect to $\ln \sigma$, which is useful for imposing bounds while guaranteeing positive conductivity values. The derivations below use the

constitutive relations introduced in Section 1.2:

$$s = i\omega \quad (2.2)$$

$$\zeta = s\mu \quad (2.3)$$

$$\zeta^v = s\mu^v \quad (2.4)$$

$$\eta = \sigma + s\varepsilon \quad (2.5)$$

$$\eta^v = \sigma^v + s\varepsilon^v \quad (2.6)$$

$$\gamma = \sqrt{\zeta \eta^v} \quad (2.7)$$

$$\bar{\gamma} = \sqrt{\eta \zeta^v} \quad (2.8)$$

$$\Gamma = \left(\frac{\eta}{\eta^v} \right)^{1/2} (\kappa^2 + \gamma^2)^{1/2} \quad (2.9)$$

$$\bar{\Gamma} = \left(\frac{\zeta}{\zeta^v} \right)^{1/2} (\kappa^2 + \bar{\gamma}^2)^{1/2} \quad (2.10)$$

We now derive explicit analytical expressions for the EM-field sensitivities in a one-dimensional isotropic or VTI layered medium.

2.2.1 Derivatives of the local reflection coefficients

The TE- and TM-mode local reflection coefficients are given by

$$r_n^{TE} = \frac{\Gamma_n - \Gamma_{n+1}}{\Gamma_n + \Gamma_{n+1}} \quad (2.11)$$

and

$$r_n^{TM} = \frac{\eta_{n+1} \Gamma_n - \eta_n \Gamma_{n+1}}{\eta_{n+1} \Gamma_n + \eta_n \Gamma_{n+1}} \quad (2.12)$$

Differentiating these expressions gives

$$\frac{\partial r_n^{TE}}{\partial \sigma_n} = \partial_{\sigma_n} \left(\frac{\Gamma_n - \Gamma_{n+1}}{\Gamma_n + \Gamma_{n+1}} \right) = \frac{\partial_{\sigma_n} \Gamma_n}{\Gamma_n + \Gamma_{n+1}} + \frac{-(\Gamma_n - \Gamma_{n+1}) (\partial_{\sigma_n} \Gamma_n)}{(\Gamma_n + \Gamma_{n+1})^2} = \frac{(\partial_{\sigma_n} \Gamma_n)(1 - r_n^{TE})}{\Gamma_n + \Gamma_{n+1}} \quad (2.13)$$

and

$$\begin{aligned} \frac{\partial r_n^{TM}}{\partial \sigma_n} &= \frac{\partial r_n^{TM}}{\partial \Gamma_n} \frac{\partial \Gamma_n}{\partial \sigma_n} + \frac{\partial r_n^{TM}}{\partial \eta_n} \frac{\partial \eta_n}{\partial \sigma_n} = \partial_{\sigma_n} \left(\frac{\eta_{n+1} \Gamma_n - \eta_n \Gamma_{n+1}}{\eta_{n+1} \Gamma_n + \eta_n \Gamma_{n+1}} \right) \\ &= \frac{\eta_{n+1} \partial_{\sigma_n} \Gamma_n - \Gamma_{n+1} - (\eta_{n+1} \Gamma_n - \eta_n \Gamma_{n+1})(\eta_{n+1} \partial_{\sigma_n} \Gamma_n + \Gamma_{n+1})}{(\eta_{n+1} \Gamma_n + \eta_n \Gamma_{n+1})^2} \\ &= \frac{\eta_{n+1} (\partial_{\sigma_n} \Gamma_n)(1 - r_n^{TM}) - \Gamma_{n+1}(1 + r_n^{TM})}{\eta_{n+1} \Gamma_n + \eta_n \Gamma_{n+1}} \end{aligned} \quad (2.14)$$

Because the properties of layer n also enter the reflection coefficient at boundary $n - 1$, the corresponding derivatives are also required:

$$\frac{\partial r_{n-1}^{TE}}{\partial \sigma_n} = \partial_{\sigma_n} \left(\frac{\Gamma_{n-1} - \Gamma_n}{\Gamma_{n-1} + \Gamma_n} \right) = \frac{-\partial_{\sigma_n} \Gamma_n}{\Gamma_{n-1} + \Gamma_n} + \frac{-(\Gamma_{n-1} - \Gamma_n) (\partial_{\sigma_n} \Gamma_n)}{(\Gamma_{n-1} + \Gamma_n)^2} = \frac{-(\partial_{\sigma_n} \Gamma_n)(1 + r_{n-1}^{TE})}{\Gamma_{n-1} + \Gamma_n} \quad (2.15)$$

and

$$\begin{aligned}
\frac{\partial r_{n-1}^{TM}}{\partial \sigma_n} &= \partial_{\sigma_n} \left(\frac{\eta_n \Gamma_{n-1} - \eta_{n-1} \Gamma_n}{\eta_n \Gamma_{n-1} + \eta_{n-1} \Gamma_n} \right) \\
&= \frac{\Gamma_{n-1} - \eta_{n-1} (\partial_{\sigma_n} \Gamma_n) - r_{n-1}^{TM} (\Gamma_{n-1} + \eta_{n-1} (\partial_{\sigma_n} \Gamma_n))}{\eta_n \Gamma_{n-1} + \eta_{n-1} \Gamma_n} \\
&= \frac{-\eta_{n-1} (\partial_{\sigma_n} \Gamma_n) (1 + r_{n-1}^{TM}) + \Gamma_{n-1} (1 - r_{n-1}^{TM})}{\eta_n \Gamma_{n-1} + \eta_{n-1} \Gamma_n}
\end{aligned} \tag{2.16}$$

where $\frac{\partial \eta_n}{\partial \sigma_n} = 1$ and

$$\begin{aligned}
\frac{\partial \Gamma_n}{\partial \sigma_n} &= \partial_{\sigma_n} ((\kappa^2 + \gamma^2)^{1/2}) = \partial_{\sigma_n} ((\kappa^2 + s\mu_n \eta_n)^{1/2}) = \partial_{\sigma_n} ((\kappa^2 + s\mu_n (\sigma_n + s\varepsilon_n))^{1/2}) \\
&= \partial_{\sigma_n} ((\kappa^2 + s\mu_n \sigma_n + s^2 \mu_n \varepsilon_n)^{1/2}) = \frac{s\mu_n}{2\Gamma_n}
\end{aligned} \tag{2.17}$$

for the isotropic case, $\eta_n = \eta_n^v$. In the VTI formulation, the corresponding derivatives of Γ and $\bar{\Gamma}$ are

$$\frac{\partial \Gamma_n}{\partial \sigma_n} = \partial_{\sigma_n} \left(\left(\frac{\eta_n}{\eta_n^v} \right)^{1/2} (\kappa^2 + \gamma^2)^{1/2} \right) = \frac{\Gamma_n}{2\eta_n} \tag{2.18}$$

$$\frac{\partial \bar{\Gamma}_n}{\partial \sigma_n} = \partial_{\sigma_n} \left(\left(\frac{\zeta_n}{\zeta_n^v} \right)^{1/2} (\kappa^2 + \bar{\gamma}^2)^{1/2} \right) = \left(\frac{\zeta_n^v}{\zeta_n} \right)^{1/2} \frac{s\mu_n^v}{2\bar{\Gamma}_n} \tag{2.19}$$

$$\frac{\partial \Gamma_n}{\partial \sigma_n^v} = \left(\frac{\eta_n^v}{\eta_n} \right)^{1/2} \frac{s\mu_n^v}{2\bar{\Gamma}_n} - \frac{\Gamma_n}{2\eta_n^v} \tag{2.20}$$

$$\frac{\partial \bar{\Gamma}_n}{\partial \sigma_n^v} = 0 \tag{2.21}$$

The final relation implies $\partial r^{TE}/\partial \sigma_n^v = 0$. Thus the TE-mode expressions depend only on the horizontal medium properties, confirming the structural asymmetry identified in Section 1.2.3.

2.2.2 Derivatives of the global reflection coefficients

The upward- and downward-looking global reflection coefficients $R_n^{TE, TM \pm}$ have the same general recursive structure for the TE and TM modes:

$$R_n^{TE, TM} = \frac{r_n^{TE, TM} + R_{n+1}^{TE, TM} \exp(-2\Gamma_{n+1} d_{n+1})}{1 + r_n^{TE, TM} R_{n+1}^{TE, TM} \exp(-2\Gamma_{n+1} d_{n+1})} \tag{2.22}$$

For a perturbation of layer n , the direct dependence enters through r_n , allowing $\partial R_n/\partial \sigma_n$ to be written as

$$\frac{\partial R_n}{\partial \sigma_n} = \frac{\partial R_n}{\partial r_n} \frac{\partial r_n}{\partial \sigma_n} \tag{2.23}$$

where the required partial derivative with respect to r_n is

$$\frac{\partial R_n}{\partial r_n} = \frac{1 - R_n R_{n+1} \exp(-2\Gamma_{n+1} d_{n+1})}{1 + r_n R_{n+1} \exp(-2\Gamma_{n+1} d_{n+1})} \tag{2.24}$$

When the recursion is continued to layer $n - 1$, the derivative $\partial R_{n-1}/\partial\sigma_n$ must be evaluated:

$$R_{n-1} = \frac{r_{n-1} + R_n \exp(-2\Gamma_n d_n)}{1 + r_{n-1} R_n \exp(-2\Gamma_n d_n)} \quad (2.25)$$

At this stage, r_{n-1} , R_n , and E_n^2 all carry a dependence on σ_n :

$$\frac{\partial R_{n-1}}{\partial\sigma_n} = \frac{\partial R_{n-1}}{\partial r_{n-1}} \frac{\partial r_{n-1}}{\partial\sigma_n} + \frac{\partial R_{n-1}}{\partial R_n} \frac{\partial R_n}{\partial\sigma_n} + \frac{\partial R_{n-1}}{\partial E_n^2} \frac{\partial E_n^2}{\partial\sigma_n} \quad (2.26)$$

with $E_n^2 = \exp(-2\Gamma_n d_n)$ and with

$$\frac{\partial R_{n-1}}{\partial r_{n-1}} = \frac{1 - R_{n-1} R_n \exp(-2\Gamma_n d_n)}{1 + r_{n-1} R_n \exp(-2\Gamma_n d_n)} \quad (2.27)$$

$$\frac{\partial R_{n-1}}{\partial R_n} = \frac{(1 - r_{n-1} R_{n-1}) \exp(-2\Gamma_n d_n)}{1 + r_{n-1} R_n \exp(-2\Gamma_n d_n)} \quad (2.28)$$

$$\frac{\partial R_{n-1}}{\partial E_n^2} = \frac{R_n(1 - r_{n-1} R_{n-1})}{1 + r_{n-1} R_n \exp(-2\Gamma_n d_n)} \quad (2.29)$$

$$\frac{\partial E_n^2}{\partial\sigma_n} = -2d_n \exp(-2\Gamma_n d_n) (\partial_{\sigma_n} \Gamma_n) \quad (2.30)$$

$\partial_{\sigma_n} \Gamma_n$ is given by Equations (2.18)–(2.21). For the downward reflection $\frac{\partial R_n}{\partial\sigma_{n+1}} = 0$.

Continuing the recursion one layer farther gives the derivative $\partial R_{n-2}/\partial\sigma_n$:

$$\frac{\partial R_{n-2}}{\partial\sigma_n} = \frac{\partial R_{n-2}}{\partial R_{n-1}} \frac{\partial R_{n-1}}{\partial\sigma_n} \quad (2.31)$$

with

$$\frac{\partial R_{n-2}}{\partial R_{n-1}} = \frac{(1 - r_{n-2} R_{n-2}) \exp(-2\Gamma_{n-1} d_{n-1})}{1 + r_{n-2} R_{n-1} \exp(-2\Gamma_{n-1} d_{n-1})} \quad (2.32)$$

Equation (2.31) shows that, for $k \leq n - 2$, the effect of the conductivity perturbation is propagated recursively through the global reflection coefficients, giving

$$\frac{\partial R_k}{\partial\sigma_n} = \frac{\partial R_k}{\partial R_{k+1}} \frac{\partial R_{k+1}}{\partial\sigma_n} \quad (2.33)$$

The general pattern is therefore to introduce the perturbation in the local reflection coefficient associated with layer n , and then propagate its effect through the stack one layer at a time via $\partial R_k/\partial R_{k+1}$. The same structure applies to other medium-parameter Jacobians, with an appropriately modified driving term; here it is specialized to conductivity.

2.2.3 Derivatives of the EM field components

The vertical electric field component is

$$\hat{G}^{ze;xe;s}(\mathbf{x}, \mathbf{x}_0, \omega) = \hat{G}^{ze;xe;s;i}(\mathbf{x} - \mathbf{x}_0, \omega) + \frac{\cos \phi}{4\pi\eta_s^v} \int_0^\infty \tilde{g}_{zh;s}^{TM} J_1(\kappa r) \kappa^2 d\kappa \quad (2.34)$$

and its derivative follows as:

$$\frac{\partial \hat{G}^{ze;xe;s}(\mathbf{x}, \mathbf{x}_0, \omega)}{\partial\sigma_n} = \frac{\partial}{\partial\sigma_n} \hat{G}^{ze;xe;s;i}(\mathbf{x} - \mathbf{x}_0, \omega) + \frac{\cos \phi}{4\pi\eta_s^v} \int_0^\infty \frac{\partial \tilde{g}_{zh;s}^{TM}}{\partial\sigma_n} J_1(\kappa r) \kappa^2 d\kappa \quad (2.35)$$

The remaining electric-field components are

$$\begin{aligned}\hat{G}^{xe;xe;s}(\mathbf{x}, \mathbf{x}_0, \omega) &= \hat{G}^{xe;xe;s;i}(\mathbf{x} - \mathbf{x}_0, \omega) + \frac{1}{8\pi} \int_0^\infty \left(\frac{\Gamma_s \tilde{g}_{hh;s}^{TM}}{\eta_s} - \frac{\zeta_s \tilde{g}_{zz;s}^{TE}}{\bar{\Gamma}_s} \right) J_0(\kappa r) \kappa d\kappa \\ &\quad - \frac{\cos(2\phi)}{8\pi} \int_0^\infty \left(\frac{\Gamma_s \tilde{g}_{hh;s}^{TM}}{\eta_s} + \frac{\zeta_s \tilde{g}_{zz;s}^{TE}}{\bar{\Gamma}_s} \right) J_2(\kappa r) \kappa d\kappa\end{aligned}\quad (2.36)$$

$$\hat{G}^{ye;xe;s}(\mathbf{x}, \mathbf{x}_0, \omega) = \hat{G}^{ye;xe;s;i}(\mathbf{x} - \mathbf{x}_0, \omega) - \frac{\sin(2\phi)}{8\pi} \int_0^\infty \left(\frac{\Gamma_s \tilde{g}_{hh;s}^{TM}}{\eta_s} + \frac{\zeta_s \tilde{g}_{zz;s}^{TE}}{\bar{\Gamma}_s} \right) J_2(\kappa r) \kappa d\kappa \quad (2.37)$$

Their derivatives are obtained in the same manner; the complete expressions are given in Section 2.2.4.

Worked example. For the isotropic case ($\sigma = \sigma^v$) and source/receiver both x -directed electric dipoles, Equation (2.36) is the field $d = \hat{G}_{xx;s}^{ee}$ whose Jacobian $J = \partial d / \partial \sigma_n$ is a standard sanity-check case: since Bessel functions and κ do not depend on σ_n , the derivative enters the integral directly and only the kernel derivatives $\partial \tilde{g}_{hh;s}^{TM} / \partial \sigma_n$, $\partial \tilde{g}_{zz;s}^{TE} / \partial \sigma_n$ (Section 2.2.4, 2.2.4) need to be found, plus, when $n = s$, the direct dependence of the Γ_s / η_s and $\zeta_s / \bar{\Gamma}_s$ prefactors on σ_s (Section 2.2.4).

2.2.4 Derivatives of the integral coefficients

The TE- and TM-mode integrals are given by

$$I_X^0 = \int_0^\infty \kappa J_0(\kappa r) g^X d\kappa \quad (2.38)$$

$$I_X^1 = \int_0^\infty J_1(\kappa r) g^X d\kappa \quad (2.39)$$

where $J_0(\kappa r)$ and $J_1(\kappa r)$ are the zero- and first-order Bessel functions and g^X are the integral kernels. The derivatives of the Hankel integrals are given by

$$\frac{\partial I_X^0}{\partial \sigma_n} = \frac{\partial}{\partial \sigma_n} \int_0^\infty \kappa J_0(\kappa r) g^X d\kappa = \int_0^\infty \kappa J_0(\kappa r) \frac{\partial g^X}{\partial \sigma_n} d\kappa \quad (2.40)$$

To differentiate the integral coefficient $\tilde{g}_{zh;s}^{TM}$ in Equation (2.34), the derivatives of the field propagators and of the factors $P_s^{u\pm}$ and $P_s^{d\pm}$ from Section 1.3 are required:

$$\tilde{g}_{zh;s}^{TM} = P_s^{u-} W_s^u - P_s^{d-} W_s^d \quad (2.41)$$

The relevant propagators and amplitude factors are

$$W_n^u = \exp[-\Gamma_n(z_n - z)] \quad (2.42)$$

$$W_n^d = \exp[-\Gamma_n(z - z_{n-1})] \quad (2.43)$$

and

$$P_s^{u\pm} = \frac{R_s^+}{M_s} \left(\exp(-\Gamma_s d^+) \pm R_s^- \exp[-\Gamma_s(d_s + d^-)] \right) \quad (2.44)$$

$$P_s^{d\pm} = \frac{R_s^-}{M_s} \left(\exp(-\Gamma_s d^-) \pm R_s^+ \exp[-\Gamma_s(d_s + d^+)] \right) \quad (2.45)$$

In these equations the factor M_s is given by

$$M_s = 1 - R_s^- R_s^+ \exp(-2\Gamma_s d_s) \quad (2.46)$$

The derivative of the integral coefficient follows from the chain rule:

$$\frac{\partial \tilde{g}_{zh;s}^{TM}}{\partial \sigma_n} = \frac{\partial \tilde{g}_{zh;s}^{TM}}{\partial P_s^{u-}} \frac{\partial P_s^{u-}}{\partial \sigma_n} + \frac{\partial \tilde{g}_{zh;s}^{TM}}{\partial W_s^u} \frac{\partial W_s^u}{\partial \sigma_n} + \frac{\partial \tilde{g}_{zh;s}^{TM}}{\partial P_s^{d-}} \frac{\partial P_s^{d-}}{\partial \sigma_n} + \frac{\partial \tilde{g}_{zh;s}^{TM}}{\partial W_s^d} \frac{\partial W_s^d}{\partial \sigma_n} \quad (2.47)$$

$$= W_s^u \frac{\partial P_s^{u-}}{\partial \sigma_n} + P_s^{u-} \frac{\partial W_s^u}{\partial \sigma_n} - W_s^d \frac{\partial P_s^{d-}}{\partial \sigma_n} - P_s^{d-} \frac{\partial W_s^d}{\partial \sigma_n} \quad (2.48)$$

where

$$\frac{\partial W_s^u}{\partial \sigma_n} = -(z_n - z) \exp[-\Gamma_n(z_n - z)] (\partial_{\sigma_n} \Gamma_n) \quad (2.49)$$

$$\frac{\partial W_s^d}{\partial \sigma_n} = -(z - z_{n-1}) \exp[-\Gamma_n(z - z_{n-1})] (\partial_{\sigma_n} \Gamma_n) \quad (2.50)$$

and

$$\frac{\partial P_s^{u\pm}}{\partial \sigma_n} = \frac{\partial P_s^{u\pm}}{\partial R_s^+} \frac{\partial R_s^+}{\partial \sigma_n} + \frac{\partial P_s^{u\pm}}{\partial R_s^-} \frac{\partial R_s^-}{\partial \sigma_n} + \frac{\partial P_s^{u\pm}}{\partial M_s} \frac{\partial M_s}{\partial \sigma_n} + \frac{\partial P_s^{u\pm}}{\partial E_1^u} \frac{\partial E_1^u}{\partial \sigma_n} + \frac{\partial P_s^{u\pm}}{\partial E_2^u} \frac{\partial E_2^u}{\partial \sigma_n} \quad (2.51)$$

The derivative $\partial P_s^{u\pm}/\partial \sigma_n$ requires the following structural partial derivatives:

$$\frac{\partial P_s^{u\pm}}{\partial R_s^+} = \frac{1}{M_s} \left(\exp(-\Gamma_s d^+) \pm R_s^- \exp[-\Gamma_s(d_s + d^-)] \right) \quad (2.52)$$

$$\frac{\partial P_s^{u\pm}}{\partial M_s} = \frac{R_s^+}{M_s^2} \left(\exp(-\Gamma_s d^+) \pm R_s^- \exp[-\Gamma_s(d_s + d^-)] \right) \quad (2.53)$$

$$\frac{\partial P_s^{u\pm}}{\partial R_s^-} = \pm \frac{R_s^+}{M_s} \exp[-\Gamma_s(d_s + d^-)] \quad (2.54)$$

$$\frac{\partial P_s^{u\pm}}{\partial E_1^u} = \frac{R_s^+}{M_s} \quad (2.55)$$

$$\frac{\partial P_s^{u\pm}}{\partial E_2^u} = \pm \frac{R_s^- R_s^+}{M_s} \quad (2.56)$$

$$\frac{\partial M_s}{\partial \sigma_n} = \frac{\partial M_s}{\partial R_s^-} \frac{\partial R_s^-}{\partial \sigma_n} + \frac{\partial M_s}{\partial R_s^+} \frac{\partial R_s^+}{\partial \sigma_n} + \frac{\partial M_s}{\partial E_3} \frac{\partial E_3}{\partial \sigma_n} \quad (2.57)$$

$$\frac{\partial M_s}{\partial R_s^\pm} = R_s^\mp \exp(-2\Gamma_s d_s) \quad (2.58)$$

$$\frac{\partial M_s}{\partial E_3} = R_s^- R_s^+ \quad (2.59)$$

$$\frac{\partial E_3}{\partial \sigma_n} = -2d_s \exp(-2\Gamma_s d_s) \partial_{\sigma_n} \Gamma_s \quad (2.60)$$

$$\frac{\partial E_1^u}{\partial \sigma_n} = -d^+ \exp(-\Gamma_s d^+) \partial_{\sigma_n} \Gamma_s \quad (2.61)$$

$$\frac{\partial E_2^u}{\partial \sigma_n} = -(d_s + d^-) \exp[-\Gamma_s(d_s + d^-)] \partial_{\sigma_n} \Gamma_s \quad (2.62)$$

The quantities $\partial R_s^\pm/\partial \sigma_n$ are obtained from the reflection-coefficient recursion derived above.

The derivative of $P_s^{d\pm}$ is obtained analogously:

$$\frac{\partial P_s^{d\pm}}{\partial \sigma_n} = \frac{\partial P_s^{d\pm}}{\partial R_s^-} \frac{\partial R_s^-}{\partial \sigma_n} + \frac{\partial P_s^{d\pm}}{\partial R_s^+} \frac{\partial R_s^+}{\partial \sigma_n} + \frac{\partial P_s^{d\pm}}{\partial M_s} \frac{\partial M_s}{\partial \sigma_n} + \frac{\partial P_s^{d\pm}}{\partial E_1^d} \frac{\partial E_1^d}{\partial \sigma_n} + \frac{\partial P_s^{d\pm}}{\partial E_2^d} \frac{\partial E_2^d}{\partial \sigma_n} \quad (2.63)$$

$$\frac{\partial P_s^{d\pm}}{\partial R_s^-} = \frac{1}{M_s} \left(\exp(-\Gamma_s d^-) \pm R_s^+ \exp[-\Gamma_s(d_s + d^+)] \right) \quad (2.64)$$

$$\frac{\partial P_s^{d\pm}}{\partial M_s} = \frac{R_s^-}{M_s^2} \left(\exp(-\Gamma_s d^-) \pm R_s^+ \exp[-\Gamma_s(d_s + d^+)] \right) \quad (2.65)$$

$$\frac{\partial P_s^{d\pm}}{\partial R_s^+} = \pm \frac{R_s^-}{M_s} \exp[-\Gamma_s(d_s + d^+)] \quad (2.66)$$

$$\frac{\partial P_s^{d\pm}}{\partial E_1^d} = \frac{R_s^-}{M_s} \quad (2.67)$$

$$\frac{\partial P_s^{d\pm}}{\partial E_2^d} = \pm \frac{R_s^+ R_s^-}{M_s} \quad (2.68)$$

$$\frac{\partial M_s}{\partial \sigma_n} = \frac{\partial M_s}{\partial R_s^+} \frac{\partial R_s^+}{\partial \sigma_n} + \frac{\partial M_s}{\partial R_s^-} \frac{\partial R_s^-}{\partial \sigma_n} + \frac{\partial M_s}{\partial E_3} \frac{\partial E_3}{\partial \sigma_n} \quad (2.69)$$

$$\frac{\partial M_s}{\partial R_s^\pm} = R_s^\mp \exp(-2\Gamma_s d_s) \quad (2.70)$$

$$\frac{\partial M_s}{\partial E_3} = R_s^- R_s^+ \quad (2.71)$$

$$\frac{\partial E_3}{\partial \sigma_n} = -2d_s \exp(-2\Gamma_s d_s) \partial_{\sigma_n} \Gamma_s \quad (2.72)$$

$$\frac{\partial E_1^d}{\partial \sigma_n} = -d^- \exp(-\Gamma_s d^-) \partial_{\sigma_n} \Gamma_s \quad (2.73)$$

$$\frac{\partial E_2^d}{\partial \sigma_n} = -(d_s + d^+) \exp[-\Gamma_s(d_s + d^+)] \partial_{\sigma_n} \Gamma_s \quad (2.74)$$

TM horizontal kernel $\tilde{g}_{hh;s}^{TM}$ and its derivatives

The horizontal TM kernel entering the x - and y -directed field components in Equations (2.36)–(2.37) is

$$\tilde{g}_{hh;s}^{TM} = P_s^{u+} W_s^u + P_s^{d+} W_s^d, \quad (2.75)$$

Here P_s^{u+} and P_s^{d+} correspond to the $\pm = +$ cases of Equations (2.44)–(2.45). This definition is consistent with the source-depth identity $\partial \tilde{g}_{zh;s}^{TM} / \partial z_0 = \Gamma_s \tilde{g}_{hh;s}^{TM}$, which lies outside the present scope. Applying the product rule gives

$$\frac{\partial \tilde{g}_{hh;s}^{TM}}{\partial \sigma_n} = W_s^u \frac{\partial P_s^{u+}}{\partial \sigma_n} + P_s^{u+} \frac{\partial W_s^u}{\partial \sigma_n} + W_s^d \frac{\partial P_s^{d+}}{\partial \sigma_n} + P_s^{d+} \frac{\partial W_s^d}{\partial \sigma_n}, \quad (2.76)$$

with $\partial W_s^{u,d} / \partial \sigma_n$ as in Equations (2.49)–(2.50). The chain rule for $\partial P_s^{u+} / \partial \sigma_n$ mirrors Equation (2.51) with $\pm = +$:

$$\frac{\partial P_s^{u+}}{\partial \sigma_n} = \frac{\partial P_s^{u+}}{\partial R_s^+} \frac{\partial R_s^+}{\partial \sigma_n} + \frac{\partial P_s^{u+}}{\partial R_s^-} \frac{\partial R_s^-}{\partial \sigma_n} + \frac{\partial P_s^{u+}}{\partial M_s} \frac{\partial M_s}{\partial \sigma_n} + \frac{\partial P_s^{u+}}{\partial E_1^u} \frac{\partial E_1^u}{\partial \sigma_n} + \frac{\partial P_s^{u+}}{\partial E_2^u} \frac{\partial E_2^u}{\partial \sigma_n}, \quad (2.77)$$

with structural partial derivatives (Equations (2.52)–(2.56) evaluated at $\pm = +$):

$$\begin{aligned}\frac{\partial P_s^{u+}}{\partial R_s^+} &= \frac{E_1^u + R_s^- E_2^u}{M_s}, & \frac{\partial P_s^{u+}}{\partial M_s} &= \frac{R_s^+ (E_1^u + R_s^- E_2^u)}{M_s^2}, \\ \frac{\partial P_s^{u+}}{\partial R_s^-} &= \frac{R_s^+ E_2^u}{M_s}, & \frac{\partial P_s^{u+}}{\partial E_1^u} &= \frac{R_s^+}{M_s}, & \frac{\partial P_s^{u+}}{\partial E_2^u} &= \frac{R_s^+ R_s^-}{M_s}.\end{aligned}\quad (2.78)$$

The derivative of P_s^{d+} is obtained in the same way from Equation (2.63), evaluated at $\pm = +$:

$$\frac{\partial P_s^{d+}}{\partial \sigma_n} = \frac{\partial P_s^{d+}}{\partial R_s^-} \frac{\partial R_s^-}{\partial \sigma_n} + \frac{\partial P_s^{d+}}{\partial R_s^+} \frac{\partial R_s^+}{\partial \sigma_n} + \frac{\partial P_s^{d+}}{\partial M_s} \frac{\partial M_s}{\partial \sigma_n} + \frac{\partial P_s^{d+}}{\partial E_1^d} \frac{\partial E_1^d}{\partial \sigma_n} + \frac{\partial P_s^{d+}}{\partial E_2^d} \frac{\partial E_2^d}{\partial \sigma_n}, \quad (2.79)$$

$$\begin{aligned}\frac{\partial P_s^{d+}}{\partial R_s^-} &= \frac{E_1^d + R_s^+ E_2^d}{M_s}, & \frac{\partial P_s^{d+}}{\partial M_s} &= \frac{R_s^- (E_1^d + R_s^+ E_2^d)}{M_s^2}, \\ \frac{\partial P_s^{d+}}{\partial R_s^+} &= \frac{R_s^- E_2^d}{M_s}, & \frac{\partial P_s^{d+}}{\partial E_1^d} &= \frac{R_s^-}{M_s}, & \frac{\partial P_s^{d+}}{\partial E_2^d} &= \frac{R_s^+ R_s^-}{M_s}.\end{aligned}\quad (2.80)$$

The derivatives $\partial M_s / \partial \sigma_n$ and $\partial E_i / \partial \sigma_n$ are identical to Equations (2.57)–(2.74).

TE kernel $\tilde{g}_{zz;s}^{TE}$ and its derivatives

The TE kernel in Equations (2.36)–(2.37) has the same algebraic structure as the TM kernel, with Γ_s replaced by $\bar{\Gamma}_s$ and TE reflection coefficients used throughout. Define the TE field propagators

$$\bar{W}_s^u = \exp[-\bar{\Gamma}_s(z_s - z)], \quad \bar{W}_s^d = \exp[-\bar{\Gamma}_s(z - z_{s-1})], \quad (2.81)$$

the TE multiple-reflection factor

$$\bar{M}_s = 1 - R_s^{TE,-} R_s^{TE,+} \exp(-2\bar{\Gamma}_s d_s), \quad (2.82)$$

the TE source-layer exponentials $\bar{E}_1^u = e^{-\bar{\Gamma}_s d^+}$, $\bar{E}_2^u = e^{-\bar{\Gamma}_s(d_s + d^-)}$, $\bar{E}_3 = e^{-2\bar{\Gamma}_s d_s}$, $\bar{E}_1^d = e^{-\bar{\Gamma}_s d^-}$, $\bar{E}_2^d = e^{-\bar{\Gamma}_s(d_s + d^+)}$, and the TE amplitude factors for the $+$ variant:

$$\bar{P}_s^{u+} = \frac{R_s^{TE,+}}{\bar{M}_s} (\bar{E}_1^u + R_s^{TE,-} \bar{E}_2^u), \quad \bar{P}_s^{d+} = \frac{R_s^{TE,-}}{\bar{M}_s} (\bar{E}_1^d + R_s^{TE,+} \bar{E}_2^d). \quad (2.83)$$

The TE kernel can then be written as

$$\tilde{g}_{zz;s}^{TE} = \bar{P}_s^{u+} \bar{W}_s^u + \bar{P}_s^{d+} \bar{W}_s^d. \quad (2.84)$$

Because $\bar{\Gamma}_n$ does not depend on σ_n^v (Equation (2.21)), the complete TE kernel is independent of vertical conductivity:

$$\frac{\partial \tilde{g}_{zz;s}^{TE}}{\partial \sigma_n^v} = 0 \quad \text{for all } n. \quad (2.85)$$

For horizontal conductivity, the product rule gives

$$\frac{\partial \tilde{g}_{zz;s}^{TE}}{\partial \sigma_{h,n}} = \bar{W}_s^u \frac{\partial \bar{P}_s^{u+}}{\partial \sigma_{h,n}} + \bar{P}_s^{u+} \frac{\partial \bar{W}_s^u}{\partial \sigma_{h,n}} + \bar{W}_s^d \frac{\partial \bar{P}_s^{d+}}{\partial \sigma_{h,n}} + \bar{P}_s^{d+} \frac{\partial \bar{W}_s^d}{\partial \sigma_{h,n}}, \quad (2.86)$$

where the propagator derivatives use $\partial\bar{\Gamma}_n/\partial\sigma_{h,n}$ from Equation (2.19):

$$\frac{\partial\bar{W}_s^u}{\partial\sigma_{h,n}} = -(z_s - z) \bar{W}_s^u \frac{\partial\bar{\Gamma}_n}{\partial\sigma_{h,n}}, \quad \frac{\partial\bar{W}_s^d}{\partial\sigma_{h,n}} = -(z - z_{s-1}) \bar{W}_s^d \frac{\partial\bar{\Gamma}_n}{\partial\sigma_{h,n}}. \quad (2.87)$$

The chain rule for $\bar{P}_s^{u+}$ follows Equation (2.77) after applying the substitutions $R_s^\pm \rightarrow R_s^{TE,\pm}$, $M_s \rightarrow \bar{M}_s$, and $E_i \rightarrow \bar{E}_i$:

$$\begin{aligned} \frac{\partial\bar{P}_s^{u+}}{\partial R_s^{TE,+}} &= \frac{\bar{E}_1^u + R_s^{TE,-} \bar{E}_2^u}{\bar{M}_s}, & \frac{\partial\bar{P}_s^{u+}}{\partial\bar{M}_s} &= \frac{R_s^{TE,+} (\bar{E}_1^u + R_s^{TE,-} \bar{E}_2^u)}{\bar{M}_s^2}, \\ \frac{\partial\bar{P}_s^{u+}}{\partial R_s^{TE,-}} &= \frac{R_s^{TE,+} \bar{E}_2^u}{\bar{M}_s}, & \frac{\partial\bar{P}_s^{u+}}{\partial\bar{E}_1^u} &= \frac{R_s^{TE,+}}{\bar{M}_s}, & \frac{\partial\bar{P}_s^{u+}}{\partial\bar{E}_2^u} &= \frac{R_s^{TE,+} R_s^{TE,-}}{\bar{M}_s}, \end{aligned} \quad (2.88)$$

$$\frac{\partial\bar{M}_s}{\partial\sigma_{h,n}} = \frac{\partial\bar{M}_s}{\partial R_s^{TE,-}} \frac{\partial R_s^{TE,-}}{\partial\sigma_{h,n}} + \frac{\partial\bar{M}_s}{\partial R_s^{TE,+}} \frac{\partial R_s^{TE,+}}{\partial\sigma_{h,n}} + \frac{\partial\bar{M}_s}{\partial\bar{E}_3} \frac{\partial\bar{E}_3}{\partial\sigma_{h,n}}, \quad (2.89)$$

$$\frac{\partial\bar{M}_s}{\partial R_s^{TE,\pm}} = R_s^{TE,\mp} \bar{E}_3, \quad \frac{\partial\bar{M}_s}{\partial\bar{E}_3} = R_s^{TE,-} R_s^{TE,+}, \quad \frac{\partial\bar{E}_3}{\partial\sigma_{h,n}} = -2d_s \bar{E}_3 \frac{\partial\bar{\Gamma}_s}{\partial\sigma_{h,n}}, \quad (2.90)$$

$$\frac{\partial\bar{E}_1^u}{\partial\sigma_{h,n}} = -d^+ \bar{E}_1^u \frac{\partial\bar{\Gamma}_s}{\partial\sigma_{h,n}}, \quad \frac{\partial\bar{E}_2^u}{\partial\sigma_{h,n}} = -(d_s + d^-) \bar{E}_2^u \frac{\partial\bar{\Gamma}_s}{\partial\sigma_{h,n}}, \quad (2.91)$$

The corresponding expressions for $\bar{P}_s^{d+}$ follow analogously from Equation (2.79):

$$\begin{aligned} \frac{\partial\bar{P}_s^{d+}}{\partial R_s^{TE,-}} &= \frac{\bar{E}_1^d + R_s^{TE,+} \bar{E}_2^d}{\bar{M}_s}, & \frac{\partial\bar{P}_s^{d+}}{\partial\bar{M}_s} &= \frac{R_s^{TE,-} (\bar{E}_1^d + R_s^{TE,+} \bar{E}_2^d)}{\bar{M}_s^2}, \\ \frac{\partial\bar{P}_s^{d+}}{\partial R_s^{TE,+}} &= \frac{R_s^{TE,-} \bar{E}_2^d}{\bar{M}_s}, & \frac{\partial\bar{P}_s^{d+}}{\partial\bar{E}_1^d} &= \frac{R_s^{TE,-}}{\bar{M}_s}, & \frac{\partial\bar{P}_s^{d+}}{\partial\bar{E}_2^d} &= \frac{R_s^{TE,-} R_s^{TE,+}}{\bar{M}_s}, \end{aligned} \quad (2.92)$$

$$\frac{\partial\bar{E}_1^d}{\partial\sigma_{h,n}} = -d^- \bar{E}_1^d \frac{\partial\bar{\Gamma}_s}{\partial\sigma_{h,n}}, \quad \frac{\partial\bar{E}_2^d}{\partial\sigma_{h,n}} = -(d_s + d^+) \bar{E}_2^d \frac{\partial\bar{\Gamma}_s}{\partial\sigma_{h,n}}. \quad (2.93)$$

The TE global-reflection derivatives $\partial R_s^{TE,\pm}/\partial\sigma_{h,n}$ follow the recursion in Section 2.2.2, with (Γ, r^{TM}, R^{TM}) replaced by $(\bar{\Gamma}, r^{TE}, R^{TE})$.

Full derivatives of the horizontal field components

The horizontal field components in Equations (2.36)–(2.37) contain products of the source-layer prefactors Γ_s/η_s and $\zeta_s/\bar{\Gamma}_s$ with their respective kernels. If $n = s$, these prefactors also depend on $\sigma_{\bullet,s}$ and therefore contribute additional derivative terms. Applying the product rule gives:

$$\frac{\partial}{\partial\sigma_{\bullet,n}} \left(\frac{\Gamma_s}{\eta_s} \tilde{g}_{hh;s}^{TM} \right) = \delta_{ns} \frac{\partial(\Gamma_s/\eta_s)}{\partial\sigma_{\bullet,s}} \tilde{g}_{hh;s}^{TM} + \frac{\Gamma_s}{\eta_s} \frac{\partial\tilde{g}_{hh;s}^{TM}}{\partial\sigma_{\bullet,n}}, \quad (2.94)$$

$$\frac{\partial}{\partial \sigma_{\bullet,n}} \left(\frac{\zeta_s}{\bar{\Gamma}_s} \tilde{g}_{zz;s}^{TE} \right) = \delta_{ns} \frac{\partial(\zeta_s/\bar{\Gamma}_s)}{\partial \sigma_{\bullet,s}} \tilde{g}_{zz;s}^{TE} + \frac{\zeta_s}{\bar{\Gamma}_s} \frac{\partial \tilde{g}_{zz;s}^{TE}}{\partial \sigma_{\bullet,n}}. \quad (2.95)$$

Because $\zeta_s = s\mu_s$ has no conductivity dependence, the non-zero prefactor derivatives reduce to the following expressions, using Equations (2.18)–(2.21):

$$\frac{\partial(\Gamma_s/\eta_s)}{\partial \sigma_{h,s}} = \frac{(\partial_{\sigma_s} \Gamma_s) \eta_s - \Gamma_s}{\eta_s^2} = \frac{\Gamma_s/2 - \Gamma_s}{\eta_s^2} = -\frac{\Gamma_s}{2\eta_s^2}, \quad (2.96)$$

$$\frac{\partial(\Gamma_s/\eta_s)}{\partial \sigma_s^v} = \frac{1}{\eta_s} \frac{\partial \Gamma_s}{\partial \sigma_s^v} = \frac{1}{\eta_s} \left[\left(\frac{\eta_s^v}{\eta_s} \right)^{1/2} \frac{s\mu_s^v}{2\bar{\Gamma}_s} - \frac{\Gamma_s}{2\eta_s^v} \right], \quad (2.97)$$

$$\frac{\partial(\zeta_s/\bar{\Gamma}_s)}{\partial \sigma_{h,s}} = -\frac{\zeta_s}{\bar{\Gamma}_s^2} \frac{\partial \bar{\Gamma}_s}{\partial \sigma_{h,s}} = -\frac{\zeta_s}{\bar{\Gamma}_s^2} \left(\frac{\zeta_s^v}{\zeta_s} \right)^{1/2} \frac{s\mu_s^v}{2\bar{\Gamma}_s}, \quad (2.98)$$

$$\frac{\partial(\zeta_s/\bar{\Gamma}_s)}{\partial \sigma_s^v} = 0. \quad (2.99)$$

Combining the preceding terms yields the complete derivatives of the horizontal field components:

$$\begin{aligned} \frac{\partial \hat{G}^{xe;xe;s}}{\partial \sigma_{\bullet,n}} &= \frac{\partial \hat{G}^{xe;xe;s;i}}{\partial \sigma_{\bullet,n}} \\ &+ \frac{1}{8\pi} \int_0^\infty \left[\delta_{ns} \frac{\partial(\Gamma_s/\eta_s)}{\partial \sigma_{\bullet,s}} \tilde{g}_{hh;s}^{TM} + \frac{\Gamma_s}{\eta_s} \frac{\partial \tilde{g}_{hh;s}^{TM}}{\partial \sigma_{\bullet,n}} - \delta_{ns} \frac{\partial(\zeta_s/\bar{\Gamma}_s)}{\partial \sigma_{\bullet,s}} \tilde{g}_{zz;s}^{TE} - \frac{\zeta_s}{\bar{\Gamma}_s} \frac{\partial \tilde{g}_{zz;s}^{TE}}{\partial \sigma_{\bullet,n}} \right] J_0(\kappa r) \kappa d\kappa \\ &- \frac{\cos(2\phi)}{8\pi} \int_0^\infty \left[\delta_{ns} \frac{\partial(\Gamma_s/\eta_s)}{\partial \sigma_{\bullet,s}} \tilde{g}_{hh;s}^{TM} + \frac{\Gamma_s}{\eta_s} \frac{\partial \tilde{g}_{hh;s}^{TM}}{\partial \sigma_{\bullet,n}} + \delta_{ns} \frac{\partial(\zeta_s/\bar{\Gamma}_s)}{\partial \sigma_{\bullet,s}} \tilde{g}_{zz;s}^{TE} + \frac{\zeta_s}{\bar{\Gamma}_s} \frac{\partial \tilde{g}_{zz;s}^{TE}}{\partial \sigma_{\bullet,n}} \right] J_2(\kappa r) \kappa d\kappa, \end{aligned} \quad (2.100)$$

$$\frac{\partial \hat{G}^{ye;ye;s}}{\partial \sigma_{\bullet,n}} = \frac{\partial \hat{G}^{ye;ye;s;i}}{\partial \sigma_{\bullet,n}} - \frac{\sin(2\phi)}{8\pi} \int_0^\infty \left[\delta_{ns} \frac{\partial(\Gamma_s/\eta_s)}{\partial \sigma_{\bullet,s}} \tilde{g}_{hh;s}^{TM} + \frac{\Gamma_s}{\eta_s} \frac{\partial \tilde{g}_{hh;s}^{TM}}{\partial \sigma_{\bullet,n}} + \delta_{ns} \frac{\partial(\zeta_s/\bar{\Gamma}_s)}{\partial \sigma_{\bullet,s}} \tilde{g}_{zz;s}^{TE} + \frac{\zeta_s}{\bar{\Gamma}_s} \frac{\partial \tilde{g}_{zz;s}^{TE}}{\partial \sigma_{\bullet,n}} \right] J_2(\kappa r) \kappa d\kappa \quad (2.101)$$

For $n \neq s$, the Kronecker-delta terms vanish and only the kernel derivatives remain. When $n = s$, the additional prefactor contributions from (2.96)–(2.99) must also be included. The required kernel derivatives are given in Section 2.2.4 and Section 2.2.4.

2.2.5 From σ -derivatives to a resistivity Jacobian

The derivatives above are expressed in terms of σ_n and σ_n^v , or equivalently η_n, η_n^v through Equations (2.5)–(2.6). If the model is instead parameterized by resistivity $\rho_n = 1/\sigma_n$ and anisotropy $\lambda_n = \sigma_n/\sigma_n^v$, the chain rule provides the conversion from conductivity to resistivity derivatives:

$$\eta_n = \frac{1}{\rho_n} + i\omega\varepsilon_0\varepsilon_n, \quad \eta_n^v = \frac{1}{\rho_n\lambda_n^2} + i\omega\varepsilon_0\varepsilon_n^v \quad \implies \quad \frac{\partial \eta_n}{\partial \rho_n} = -\frac{1}{\rho_n^2}, \quad \frac{\partial \eta_n^v}{\partial \rho_n} = -\frac{1}{\rho_n^2\lambda_n^2}. \quad (2.102)$$

An implementation based directly on the eta/zeta seeds can therefore use $\partial \eta_n/\partial \rho_n$ and $\partial \eta_n^v/\partial \rho_n$, together with the conductivity derivatives derived above, in the chain rule. In particular, $\partial F/\partial \rho_n$ is obtained by combining the horizontal and vertical conductivity derivatives with $\partial \sigma_n/\partial \rho_n = -1/\rho_n^2$ and $\partial \sigma_n^v/\partial \rho_n = -1/(\rho_n^2\lambda_n^2)$ at fixed λ_n .

Chapter 3

Other gradients

If you are interested in the derivations for the other parameters, or in discussing potential applications, extensions, or collaboration, please feel free to reach out to me at www.wouterdeleersnyder.be.

- 3.1 Gradients with respect to the anisotropy factor (aniso)**
- 3.2 Gradients with respect to electric permittivity (epermH, epermV)**
- 3.3 Gradients with respect to magnetic permeability (mpermH, mpermV)**
- 3.4 Gradients with respect to interface depths (depth, thickness)**
- 3.5 Gradients with respect to source/receiver depth**
- 3.6 Gradients with respect to offset (off)**
- 3.7 Gradients with respect to frequency (freq)**
- 3.8 Gradients with respect to horizontal source position**
- 3.9 Gradients with respect to source/receiver orientation**

Appendix A

Notation

A.1 Layer/medium parameters

Symbol	Meaning	Units/notes
ρ_n	Horizontal resistivity	Ωm ; $\rho_n = 1/\sigma_n$
$\lambda_n = \sigma_n/\sigma_n^v$	Anisotropy factor	$= \sqrt{\rho_n^v/\rho_n} \geq 1$
$\varepsilon_n, \varepsilon_n^v$	Relative electric permittivity	horizontal/vertical
μ_n, μ_n^v	Relative magnetic permeability	horizontal/vertical
$\sigma_n = 1/\rho_n, \sigma_n^v$	Horizontal/vertical conductivity	S/m
η_n, η_n^v	Admittivity	$= \sigma + i\omega\varepsilon_0\varepsilon$
ζ_n, ζ_n^v	Impedivity	$= i\omega\mu_0\mu$
z_m	Interface depth	
$d_m = z_m - z_{m-1}$	Layer thickness	finite for interior layers
$\omega = 2\pi f$	Angular frequency	
$s = i\omega$	Laplace/frequency variable	$e^{+i\omega t}$ convention
ε_0, μ_0	Vacuum permittivity/permeability	

A.2 Propagation, reflection and kernel quantities

Symbol	Meaning
κ	Hankel-transform wavenumber
Γ_n	TM propagation constant (Section 1.2.3)
$\bar{\Gamma}_n$	TE propagation constant
$\gamma_n = \sqrt{\zeta_n\eta_n^v}, \bar{\gamma}_n = \sqrt{\eta_n\zeta_n^v}$	intermediate quantities
r_n^{TE}, r_n^{TM}	Local reflection coefficients (Section 1.3)
R_n^{TE}, R_n^{TM}	Global reflection coefficients
$W_s^{u,d}, \bar{W}_s^{u,d}$	Source-layer field propagators
$P_s^{u,d\pm}, \bar{P}_s^{u,d\pm}$	TM/TE amplitude factors
$M_s, \bar{M}_s$	Source-layer multiple factor
$E_1^{u,d}, E_2^{u,d}, E_3$	Source-layer exponentials
$\tilde{g}_{zh;s}^{TM}, \tilde{g}_{hh;s}^{TM}, \tilde{g}_{zz;s}^{TE}$	Integral (wavenumber-domain) kernels
ϕ	Source-receiver azimuth

Symbol	Meaning
r	Source-receiver radial offset
z_0^{src}, z	Source/receiver depth
d^+, d^-	Source-to-interface distances